IDL Assignment 0 - Group 188

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Code repository available at GitHub.

I. Introduction

This report will investigate the MNIST classification task using dimensionality reduction methods, nearest-Mean and k-NearestNeighbors (kNN), and a multiclass perceptron (MCP) with a softmax layer used for loss computation. Evaluations of the MCP will include loss in the training set and accuracy in both the training set and the test set. The datasets used are subsets of the MNIST dataset.

II. Methods

The nearest-Mean classifier is built using the mean of the training set centers per label.

PCA, t-SNE, and UMAP were used with their default parameters. Refer to [1], [2] for specific details on algorithm parameters.

The MCP has the following architecture: input layer ($D_{in} = 256$), softmax over outputs to compute multiclass cross entropy loss, inference is done by argmaxing over the output logits ($D_{out} = 10$ output logits corresponding to the output classes). The biases are initialized as $\mathbf{1}$ and the weights are sampled from a normal distribution with $\mu = 0, \sigma = 4/D_{in} + D_{out}$. The weights are optimized using gradient descent over the entire dataset. Both normalized and non-normalized loss gradients are explored. To evaluate MCP, three different values are evaluated for each hyperparameter (number of epochs $\in \{10, 50, 100, 1000\}$, learning rate (lr) $\in \{0.001, 0.01, 0.05\}$).

III. Results

The classifier has no pair of centers that have a low euclidean distance. The closest pairs of numbers are 7 and 9, 8 and 9, and 5 and 9. When we consider these characters, there is some intuition as to why their centers have a lower euclidean distance than other pairs (much of the characters could be said to overlap). From the distance matrix alone, this classifier could produce a good solution, there do not seem to be two characters which are so close that they would be prone to incorrect classification.

t-SNE and UMAP both produce topologies in which the data points form distinct clusters per label. The PCA plots less; however, the same label points inhabit the same regions (see Fig. 1). All three dimensionality reduction methods show that the centers of the different label clusters are spaced sufficiently far apart to assume nearest-Mean to be a reliable method for classification.

Fig. 4 show that characters 9 and 6 are most often misclassified in the train set. The misclassification of 9 and 7

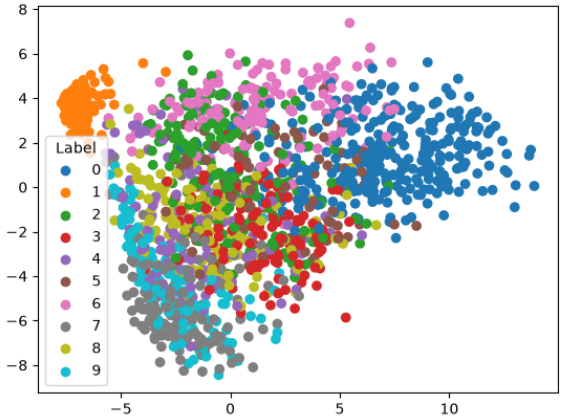


Fig. 1: PCA visualization of MNIST digit embeddings.

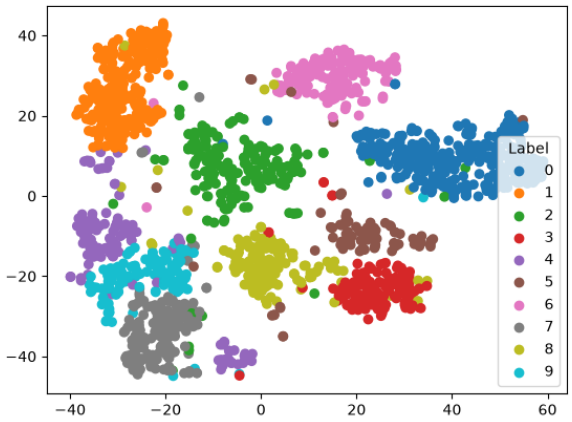


Fig. 2: t-SNE visualization of MNIST digit embeddings.

corroborates our previous analysis. 6 is most often misclassified as 0. In the test set, 6 is still most often misclassified as 0; it is also misclassified the most in both Fig. 4 and Fig. 5. kNN rarely misclassifies any character and performs really well on both the train and test sets. kNN outperforms our nearest Mean implementation by 10 percentage points on both train and test evaluations (see Table II).

Interestingly, kNN performs as well as our MCP given specific hyperparameter values (see Table III values marked in bold). We manipulate the number of epochs and the learning rate. In this experimental setup, setting the learning rate to 0.05

TABLE I: Euclidean distance matrix for all label centers.

	0	1	2	3	4	5	6	7	8	9
0	0	14.45	9.33	9.14	10.77	7.52	8.15	11.86	9.91	11.49
1	14.45	0	10.13	11.73	10.17	11.12	10.61	10.74	10.09	9.93
2	9.33	10.13	0	8.18	7.93	7.91	7.33	8.87	7.08	8.89
3	9.14	11.73	8.18	0	9.09	6.12	9.3	8.92	7.02	8.35
4	10.77	10.17	7.93	9.09	0	8	8.78	7.58	7.38	6.01
5	7.52	11.12	7.91	6.12	8	0	6.7	9.21	6.97	8.26
6	8.15	10.61	7.33	9.3	8.78	6.7	0	10.89	8.59	10.44
7	11.86	10.74	8.87	8.92	7.58	9.21	10.89	0	8.47	5.43
8	9.91	10.09	7.08	7.02	7.38	6.97	8.59	8.47	0	6.4
9	11.49	9.93	8.89	8.35	6.01	8.26	10.44	5.43	6.4	0

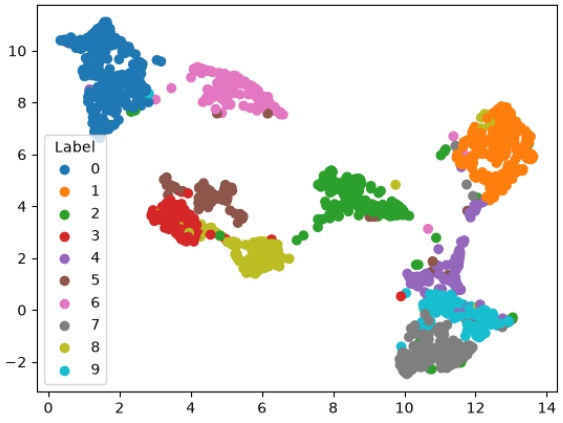


Fig. 3: UMAP visualization of MNIST digit embeddings.

TABLE II: Classification performance comparison.

Model	Train	Test
Nearest Mean	0.8635	0.8040
kNN	0.9660	0.9080

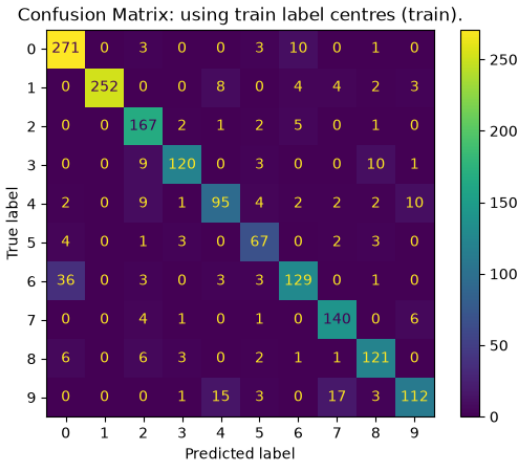


Fig. 4: Confusion matrix for nearest-Mean classifier on the training set.

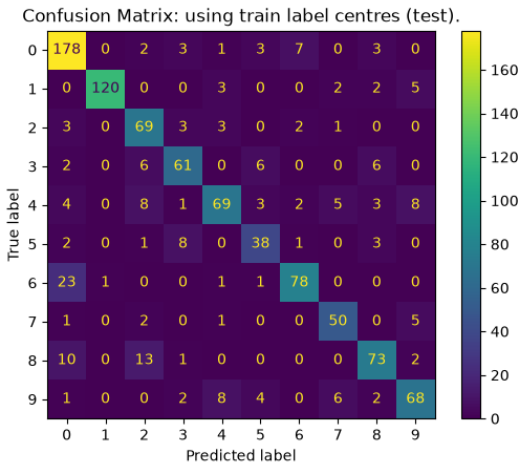


Fig. 5: Confusion matrix for nearest-Mean classifier on the test set.

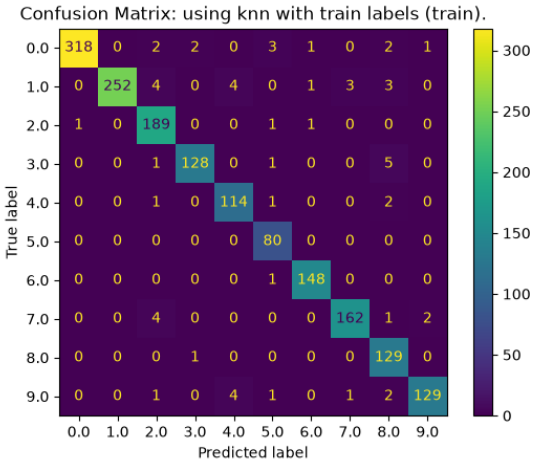


Fig. 6: Confusion matrix for kNN classifier on the training set.

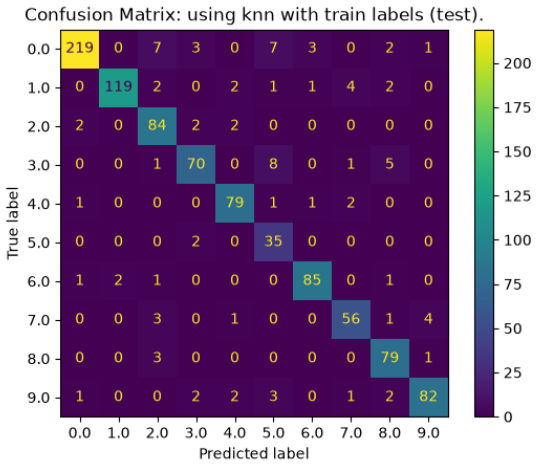


Fig. 7: Confusion matrix for kNN classifier on the test set.

TABLE III: Mean and std of train loss and test and train accuracy across 5 seeds for our MCP with normalized loss gradient.

Epochs	lr	Train Loss	Train Accuracy	Test Accuracy
10	0.001	2.26 ± 0.04	0.18 ± 0.10	0.18 ± 0.11
10	0.010	1.88 ± 0.02	0.43 ± 0.02	0.42 ± 0.02
10	0.050	1.17 ± 0.01	0.77 ± 0.00	0.73 ± 0.00
50	0.001	2.05 ± 0.03	0.40 ± 0.01	0.38 ± 0.01
50	0.010	1.14 ± 0.01	0.78 ± 0.01	0.73 ± 0.01
50	0.050	0.48 ± 0.00	0.91 ± 0.00	0.85 ± 0.00
100	0.001	1.85 ± 0.02	0.45 ± 0.02	0.43 ± 0.02
100	0.010	0.80 ± 0.00	0.86 ± 0.00	0.80 ± 0.01
100	0.050	0.33 ± 0.00	0.94 ± 0.00	0.87 ± 0.00
1000	0.001	0.79 ± 0.00	0.86 ± 0.00	0.80 ± 0.01
1000	0.010	0.24 ± 0.00	0.95 ± 0.00	0.88 ± 0.00
1000	0.050	0.11 ± 0.00	$\mathbf{0.98} \pm 0.00$	$\mathbf{0.90} \pm 0.00$

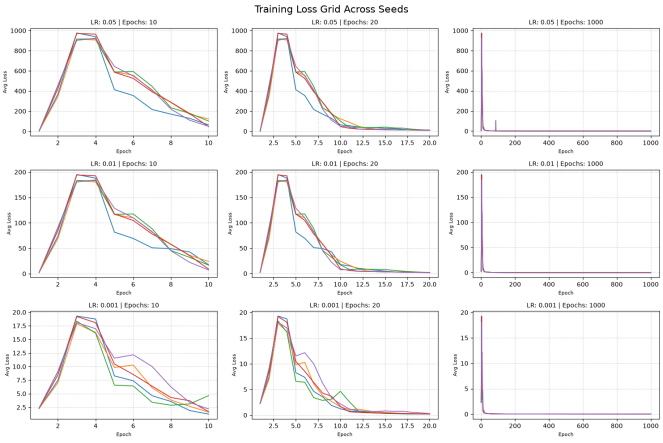


Fig. 8: Non-normalized loss grid across epochs. Epochs increase incrementally towards the right, learning rate increases downwards.

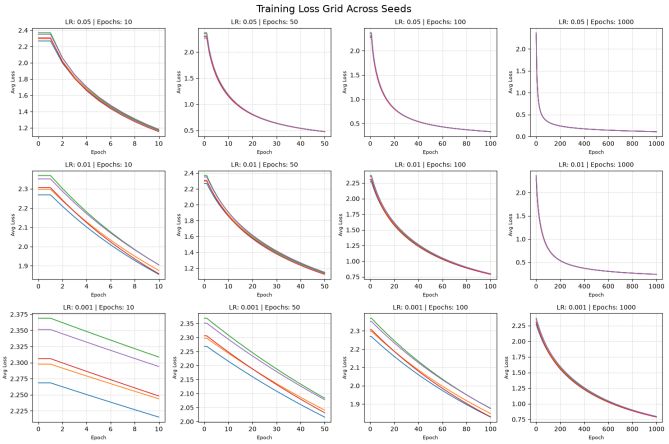


Fig. 9: Normalized loss grid showing smooth convergence. Epochs increase incrementally towards the right, learning rate increases downwards.

and the number of epochs to 1000 yielded the best accuracy and loss results. As can be seen, with increasing epochs, the gap between train and test accuracy increases as well; the model is fitting to the data. Similarly, increasing the learning rate also widens this train-test performance gap.

Additionally, the non-normalized loss curves show that the loss really explodes upward as the loss landscape is being navigated in the early epochs (Fig. 8). Using normalized gradient updates, the curve is much smoother toward convergence for all hyperparameter settings (Fig. 9).

IV. Conclusion

This set of experiments revealed that simple classifiers and dimensionality reduction methods can perform very well in image classification of the MNIST dataset. The MCP performed at its best as good as the kNN classifier. Given an increased budget of epochs, MCP could exceed the performance of these methods, however, the gains are marginal. Other methods, such as SVMs, might be more suited to solve simple classification objectives.

References

[1] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[2] T. Sainburg, L. McInnes, and T. Q. Gentner, "Parametric umap embeddings for representation and semisupervised learning," *Neural Computation*, vol. 33, no. 11, pp. 2881–2907, 2021.